Dreissena polymorpha
(zebra mussel)
Mollusks-Bivalves
Exotic

Collection Info
Point Map
Species Profile
Animated Map
Impacts

Image 1 of 3

Amy Benson - U.S. Geological Survey

Dreissena polymorpha (Pallas, 1771)

Common name: zebra mussel

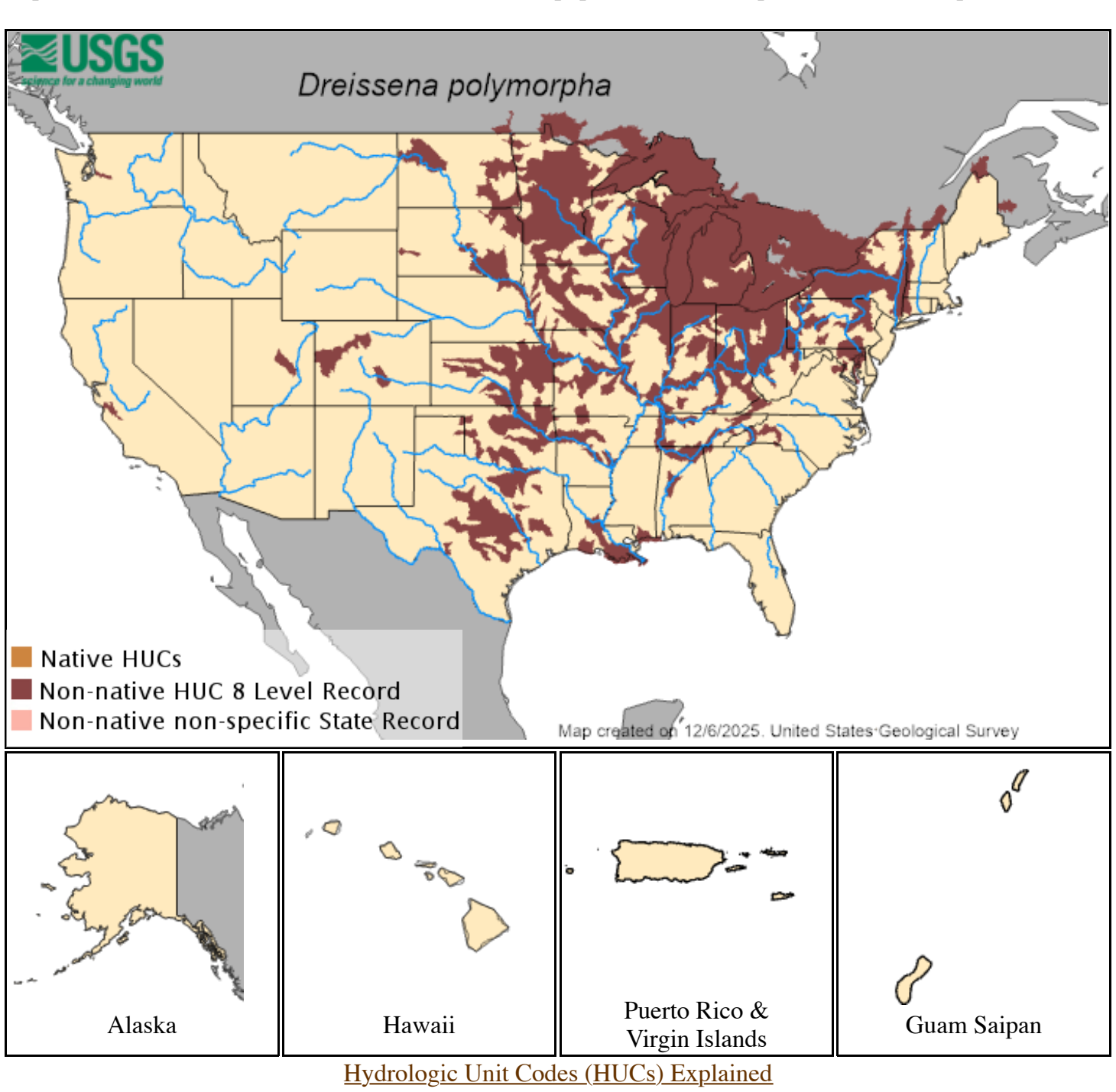
Taxonomy: available through ITIS

Injurious: This species is listed by the U.S. Fish and Wildlife Service as injurious wildlife.

Identification: The zebra mussel is a small shellfish named for the striped pattern of its shell. However, color patterns can vary to the point of having only dark or light colored shells with no stripes. This mussel is typically found attached to objects, surfaces, or other mussels by threads extending from underneath the shells. Although similar in appearance to the quagga mussel (*Dreissena bugensis*), the two species can be distinguished by their shell morphology. When placed on a surface, zebra mussels are stable on their flattened underside while quagga mussels, lacking a flat underside, will fall over. When both zebra and quagga mussels occur in the same area, differentiation can be difficult due to the phenotypic plasticity seen in quagga mussels, and thus genetic identification is necessary at times (Kerambrun et al. 2018, Beggel et al. 2015). See Mackie and Schlosser (1996) and Ram et al. (2012) for a key to adult dreissenids.

Size: adults < 50 mm, veliger (planktonic larva) 70-200 μm

Native Range: The zebra mussels is native to the Black, Caspian, and Azov Seas. In 1769, Pallas first described populations of this species from the Caspian Sea and Ural River.



USGS
science for a changing world
Dreissena polymorpha
Native HUCs
Non-native HUC 8 Level Record
Non-native non-specific State Record
Map created on 12/6/2025. United States Geological Survey
Alaska
Hawaii
Puerto Rico & Virgin Islands
Guam Saipan

Hydrologic Unit Codes (HUCs) Explained
Interactive maps: Point Distribution Maps

Nonindigenous Occurrences:

Table 1. States with nonindigenous occurrences, the earliest and latest observations in each state, and the tally and names of HUCs with observations†. Names and dates are hyperlinked to their relevant specimen records. The list of references for all nonindigenous occurrences of *Dreissena polymorpha* are found here.

State	First Observed	Last Observed	Total HUCs with observations†	HUCs with observations†
AL	1992	2023	5	Guntersville Lake; Lower Black Warrior; Pickwick Lake; Upper Black Warrior; Wheeler Lake
AR	1992	2020	12	Bull Shoals Lake; Dardanelle Reservoir; Frog-Mulberry; Lake Conway-Point Remove; Little Red; Lower Arkansas; Lower Arkansas-Maumelle; Lower Mississippi-Greenville; Lower Mississippi-Helena; Lower Mississippi-Memphis; Lower White; Robert S. Kerr Reservoir
CA	2008	2012	1	Pajaro
CO	2007	2025	4	Colorado Headwaters; Colorado Headwaters-Plateau; Lower Gunnison; Upper Arkansas
CT	1998	2022	1	Housatonic
IL	1989	2023	29	Apple-Plum; Bear-Wyaconda; Cahokia-Joachim; Chicago; Copperas-Duck; Des Plaines; Flint-Henderson; Highland-Pigeon; Iroquois; Kankakee; Lake Michigan; Little Calumet-Galien; Lower Fox; Lower Illinois; Lower Illinois-Lake Chautauqua; Lower Illinois-Senachwine Lake; Lower Ohio; Lower Ohio-Bay; Lower Rock; Lower Wabash; Peruque-Piasa; Pike-Root; Saline; The Sny; Upper Fox; Upper Illinois; Upper Mississippi-Cape Girardeau; Vermilion; Vermilion
IN	1988	2024	19	Blue-Sinking; Driftwood; Eel; Highland-Pigeon; Kankakee; Lake Michigan; Little Calumet-Galien; Lower Ohio-Little Pigeon; Lower Wabash; Middle Ohio-Laughery; Middle Wabash-Busseron; Middle Wabash-Little Vermilion; Silver-Little Kentucky; St. Joseph; St. Joseph; Tippecanoe; Upper Wabash; Upper White; Whitewater
IA	1992	2024	19	Apple-Plum; Big Papillion-Mosquito; Blackbird-Soldier; Blue Earth; Boone; Coon-Yellow; Copperas-Duck; Flint-Henderson; Grant-Little Maquoketa; Lake Red Rock; Little Sioux; Lower Des Moines; Maquoketa; Middle Cedar; Middle Des Moines; North Raccoon; Upper Chariton; Upper Iowa; Winnebago
KS	2001	2025	25	Chikaskia; Delaware; Gar-Peace; Independence-Sugar; Lower Big Blue; Lower Cottonwood; Lower Kansas, Kansas; Lower Marais Des Cygnes; Lower Republican; Lower Smoky Hill; Lower Walnut Creek; Lower Walnut River; Middle Arkansas-Slate; Middle Kansas; Middle Smoky Hill; Neosho Headwaters; Ninnescah; North Fork Ninnescah; Solomon; Upper Cottonwood; Upper Marais Des Cygnes; Upper Neosho; Upper Saline; Upper Smoky Hill; Upper Walnut River
KY	1991	2025	20	Big Sandy; Blue-Sinking; Highland-Pigeon; Kentucky Lake; Licking; Little Scioto-Tygarts; Lower Cumberland; Lower Kentucky; Lower Levisa; Lower Mississippi-Memphis; Lower Ohio; Lower Ohio-Bay; Lower Ohio-Little Pigeon; Middle Green; Middle Ohio-Laughery; Ohio Brush-Whiteoak; Saline; Silver-Little Kentucky; Upper Cumberland-Lake Cumberland; Upper Levisa
LA	1992	2022	10	Atchafalaya; Bayou Teche; East Central Louisiana Coastal; Lower Mississippi-Baton Rouge; Lower Mississippi-Greenville; Lower Mississippi-Natchez; Lower Mississippi-New Orleans; Lower Red; Mermentau; West Central Louisiana Coastal
MD	2008	2018	5	Chester-Sassafras; Gunpowder-Patapsco; Lower Susquehanna; Monocacy; Upper Chesapeake Bay
MA	2009	2015	1	Housatonic
MI	1988	2023	50	Au Gres-Rifle; Au Sable; Betsie-Platte; Betsy-Chocolay; Black; Black-Macatawa; Boardman-Charlevoix; Brule; Carp-Pine; Cedar-Ford; Cheboygan; Clinton; Detroit; Escanaba; Fishdam-Sturgeon; Flint; Huron; Kalamazoo; Kawkawlin-Pine; Lake Erie; Lake Huron; Lake Michigan; Lake St. Clair; Lake Superior; Lone Lake-Ocqueoc; Lower Grand; Manistee; Manistique River; Maple; Menominee; Millecoquins Lake-Brevoort River; Muskegon; Ontonagon; Ottawa-Stony; Pere Marquette-White; Pigeon-Wiscoggin; Pine; Raisin; Saginaw; Shiawassee; St. Clair; St. Joseph; St. Joseph; St. Marys; Tacoosh-Whitefish; Tahquamenon; Thunder Bay; Tiffin; Tittabawassee; Upper Grand
MN	1989	2024	44	Baptism-Brule; Big Fork; Buffalo-Whitewater; Chippewa; Clearwater; Clearwater-Elk; Coon-Yellow; Cottonwood; Crow; Crow Wing; Des Moines Headwaters; Eastern Wild Rice; Elk-Nokasippi; Hawk-Yellow Medicine; Kettle; La Crosse-Pine; Lake of the Woods; Lake Superior; Leech Lake; Little Fork; Long Prairie; Lower Minnesota; Lower St. Croix; Middle Minnesota; Middle Red; Mississippi Headwaters; Otter Tail; Pine; Platte-Spunk; Pomme De Terre; Prairie-Willow; Rainy Lake; Red Lakes; Redeye; Rum; Rush-Vermillion; Sandhill-Wilson; Sauk; South Fork Crow; St. Louis; Twin Cities; Upper Minnesota; Upper Red; Zumbro
MS	1992	2002	3	Lower Mississippi-Greenville; Lower Mississippi-Natchez; Mississippi Coastal
MO	1992	2019	16	Bear-Wyaconda; Bull Shoals Lake; Cahokia-Joachim; Harry S. Truman Reservoir; Lake of the Ozarks; Lower Mississippi-Memphis; Lower Missouri; Lower Missouri-Crooked; Lower Missouri-Moreau; Lower Osage; Meramec; Niangua; Peruque-Piasa; Platte; The Sny; Upper Mississippi-Cape Girardeau
NE	2006	2023	5	Big Papillion-Mosquito; Blackbird-Soldier; Keg-Weeping Water; Lewis and Clark Lake; Tarkio-Wolf
NY	1989	2024	33	Buffalo-Eighteenmile; Cattaraugus; Chaumont-Perch; Chemung; Chenango; Conewango; French; Grass; Headwaters St. Lawrence River; Hudson-Hoosic; Hudson-Wappinger; Indian; Irondequoit-Ninemile; Lake Champlain; Lake Erie; Lake Ontario; Lower Genesee; Lower Hudson; Mettawee River; Middle Hudson; Mohawk; Niagara River; Oak Orchard-Twelvemile; Oneida; Oswego; Owego-Wappasening; Raisin River-St. Lawrence River; Richelieu River; Salmon-Sandy; Schoharie; Seneca; Upper Genesee; Upper Susquehanna
NC	2023	2023	1	Upper Catawba
ND	2010	2025	9	Goose; Grand Marais-Red; Lake Sakakawea; Middle Red; Middle Sheyenne; Sandhill-Wilson; Upper James; Upper Red; Western Wild Rice
OH	1988	2023	30	Ashtabula-Chagrin; Auglaize; Black-Rocky; Blanchard; Cedar-Portage; Cuyahoga; Grand; Hocking; Huron-Vermilion; Lake Erie; Little Miami; Little Muskingum-Middle Island; Little Scioto-Tygarts; Lower Great Miami, Indiana, Ohio; Lower Maumee; Lower Scioto; Mahoning; Middle Ohio-Laughery; Mohican; Muskingum; Ohio Brush-Whiteoak; Raccoon-Symmes; Sandusky; Tiffin; Tuscarawas; Upper Ohio; Upper Ohio-Shade; Upper Ohio-Wheeling; Upper Scioto; Wills
OK	1993	2024	23	Bird; Black Bear-Red Rock; Bois D'arc-Island; Cache; Dirty-Greenleaf; Kaw Lake; Lake O' The Cherokees; Lake Texoma; Lower Canadian; Lower Cimarron; Lower Cimarron-Skeleton; Lower Neosho; Lower North Canadian; Lower North Fork Red; Lower Verdigris; Lower Washita; Lower Wolf; Middle Verdigris; Middle Washita; Northern Beaver; Polecat-Snake; Robert S. Kerr Reservoir; Washita Headwaters
PA	1989	2025	18	Beaver; Conewango; French; Lake Erie; Lehigh; Lower Allegheny; Lower Monongahela; Lower Susquehanna; Lower Susquehanna-Penns; Middle Allegheny-Redbank; Middle Allegheny-Tionesta; Raystown; Tioga; Upper Ohio; Upper Susquehanna; Upper Susquehanna-Lackawanna; Upper Susquehanna-Tunkhannock; Upper West Branch Susquehanna
SD	2014	2024	8	Fort Randall Reservoir; Lac Qui Parle; Lewis and Clark Lake; Lower James; Rapid; Upper Big Sioux; Upper James; Upper Minnesota
TN	1992	2025	12	Guntersville Lake; Holston; Kentucky Lake; Lower Cumberland; Lower Cumberland-Old Hickory Lake; Lower Cumberland-Sycamore; Lower Mississippi-Memphis; Lower Tennessee-Beech; Middle Tennessee-Chickamauga; Upper Clinch, Tennessee, Virginia; Upper French Broad; Watts Bar Lake
TX	2009	2025	31	Austin-Travis Lakes; Bois D'arc-Island; Bosque; Buchanan-Lyndon B. Johnson Lakes; Cowhouse; Denton; East Fork Trinity; Elm Fork Trinity; Jim Ned; Lake Fork; Lake Texoma; Lampasas; Leon; Little; Llano; Lower Brazos-Little Brazos; Lower Colorado-Cummins; Lower Devils; Lower Trinity-Kickapoo; Lower West Fork Trinity; Medina; Middle Colorado; Middle Guadalupe; Pecan Bayou; Richland; San Gabriel; San Saba; South Concho; Upper Clear Fork Brazos; Upper Guadalupe; Upper West Fork Trinity
UT	2008	2008	1	San Rafael
VT	1993	2019	3	Lake Champlain; Mettawee River; Otter Creek
VA	2002	2002	1	Middle Potomac-Anacostia-Occoquan
WA	2024	2024	1	Duwamish
WV	1992	2022	9	Little Muskingum-Middle Island; Lower Kanawha; Raccoon-Symmes; Tygart Valley; Upper Kanawha; Upper Monongahela; Upper Ohio; Upper Ohio-Shade; Upper Ohio-Wheeling
WI	1989	2023	32	Apple-Plum; Beartrap-Nemadji; Buffalo-Whitewater; Castle Rock; Coon-Yellow; Des Plaines; Door-Kewaunee; Duck-Pensaukee; Flambeau; Grant-Little Maquoketa; La Crosse-Pine; Lake Michigan; Lake Superior; Lake Winnebago; Lower Fox; Lower St. Croix; Lower Wisconsin; Manitowoc-Sheboygan; Menominee; Middle Rock; Milwaukee; Namekagon; Oconto; Peshtigo; Pike-Root; Rush-Vermillion; St. Louis; Upper Fox; Upper Fox; Upper Rock; Upper Wisconsin; Wolf

Table last updated 12/8/2025

† Populations may not be currently present.

Ecology: The life history of zebra mussels differs greatly from most endemic Great Lakes-region bivalves (Pennak 1989, Mackie and Schlosser 1996). Exotic dreissenids are dioecious, with fertilization occurring in the water column. Endemic bivalves are monoecious, dioecious or hermaphroditic, and some are internally fertilized by filtering sperm from the water column. Under natural thermal regimes, zebra mussel oogenesis occurs in autumn, with eggs developing until release and fertilization in spring. In thermally polluted areas, reproduction can occur continually through the year. Females generally reproduce in their second year. Eggs are expelled by the females and fertilized outside the body by the males; this process usually occurs in the spring or summer, depending on water temperature. Spawning may start when the water temperature reaches 12°C and release rate is maximized above 17-18°C (McMahon 1996). Over 40,000 eggs can be laid in a reproductive cycle and up to one million in a spawning season. Spawning may last longer in waters that are warm throughout the year.

After the eggs are fertilized, the larvae (veligers) emerge within 3 to 5 days and are free-swimming for up to a month. Optimal temperature for larval development is 20–22°C. Dispersal of larvae is normally passive by being carried with water currents. The dispersion of zebra mussels within a lake is controlled by physical conditions including wind strength, lake/shore morphometry, and current patterns (Stanczykowska and Lewandowski 1993). These conditions affect both spatial patterns of pelagic veliger density and benthic adult dispersion. Once the veliger undergoes morphological changes including development of the siphon, foot, organ systems and blood, it is known as a postveliger. Further subdivision of the larval stage has been delineated: (veliger) preshell, straight-hinged, umbonal, (postveliger) pediveliger, plantigrade, and (juvenile) settling stage (ZMIS 1996). The settling stage attaches to a substrate via protienaceous threads secreted from the byssal gland. The vast majority of veliger mortality (99%) occurs at this stage due to settlement onto unsuitable substrates. Sensitivity to changes in temperature and oxygen are also greatest at this stage. The larvae begin their juvenile stage by settling to the bottom where they crawl about on the bottom by means of a foot, searching for suitable substrate. They then attach themselves to it by means of byssal threads. Although the juveniles prefer a hard or rocky substrate, they have been known to attach to vegetation. As adults, mussels have a difficult time staying attached when water velocities exceed two meters per second.

Zebra mussels attach to any stable substrate in the water column or benthos, including rock, macrophytes, artificial surfaces (cement, steel, rope, etc.), crayfish, unionid clams, and each other, forming dense colonies called druses. Long-term stability of substrate affects population density and age distributions on those substrates. Within Polish lakes, perennial plants maintained larger populations than did annuals (Stanczykowska and Lewandowski 1993). Populations on plants also were dominated by mussels less than a year old, as compared with benthic populations. These populations of small individuals allow higher densities on plants. In areas where hard substrates are lacking, such as a mud or sand, zebra mussels cluster on any hard surface available. Given a choice of hard substrates, mussels prefer dark, rough substrates that are above the bottom of the lake bed (Kobak 2013). They also respond to the presence of predators by using byssal threads to attach more strongly to the substrate, forming aggregations, and reducing their upward movement (Kobak 2013). Research on Danish lakes shows factors that cause substrate to be unsuitable for both initial and long term colonization, including extensive siltation, some sessile benthic macroinvertebrates, macroalgae, and fluctuating water levels exposing mussels to desiccation (Smit et al. 1993). Population density of benthic adults has been observed to vary as widely as two orders of magnitude (e.g., <100 to >1500 individuals/m2) within individual Polish lakes due to these physical conditions. Tolerance limits of physical and chemical parameters are well known (Sprung 1993, Vinogradov et al. 1993, McMahon 1996).

Discrepancy exists when comparing temperature tolerance limits of North American and European populations, potentially due to the American population being founded by mussels from the southern limit of the European population's range. Most work in Europe has been done in the northern range. North American populations are generally adapted to warmer temperature regimes than their European counterparts. Although shell growth has been reported to occur at temperatures as low as 3°C, Lake St. Clair populations and some European populations display shell growth at 6–8°C. The optimal temperature range for adults extends to 20–25°C, but *D. polymorpha* can persist in temperatures up to 30°C. Short term tolerance of temperatures up to 35°C is possible if the mussels were previously acclimated to high temperatures. Rapid warming of shallow lakes has been hypothesized to detrimentally affect reproductive rates in Danish populations (Smit et al. 1993).

Oxygen demands are similar to those of other freshwater bivalves including unionids. Tolerance of "anaerobic" conditions has been reported for short time periods under certain temperatures and sizes, but zebra mussels cannot persist in hypoxic conditions. The lower limit of pO2 tolerance is 32–40 Torr at 25°C. Zebra mussels have been found in the hypolimnetic zone of lakes with oxygen levels of 0.1-11.2 mg/l, and in the epilimnetic zone with oxygen levels of 4.2–13.3 mg/l. Zebra mussels are described as poor O2 regulators, possibly explaining their low success rate in colonizing eutrophic lakes and the hypolimnion. Indeed, the distribution of Dreissenid mussels is severely limited in the central basin of Lake Erie, which routinely experiences bottom hypoxia (Karatayev et al. 2017).

The salinity tolerance of zebra mussels is low. Although some populations of European zebra mussels can be found in estuaries, their persistence has been speculatively attributed to reduced tidal fluctuation. Upper limits of freshwater bivalve salinity tolerance reach 8–10 ‰, and populations of European zebra mussels have been found to tolerate and range of salinities, from 0.6 ‰ (Rhine River) to 10.2 ‰ (Caspian Sea). North American populations generally tolerate salinity up to 4 ‰. Calcium and pH levels also influence survival and growth. In European populations, calcium concentrations of 24 mg Ca2+/l allow only 10% larval survival due to inhibition of shell development. Optimal calcium concentrations ranges from 40–55 mg Ca2+/l, but North American populations have been found in lakes with lower concentrations. North American populations require 10 mg Ca2+/l to initiate shell growth and 25 mg Ca2+/l to maintain shell growth. Larval development is inhibited at pH of 7.4. Higher rates of adult survival occur at a pH of 7.0–7.5, but populations have been found in the hypolimnetic zone of lakes with a pH of 6.6–8.0, and in the epilimnetic zone with a pH of 7.7–8.5. Optimal larval survival occurs at a pH of 8.4, and optimal adult growth occurs at pH 7.4–8.0.

The life span of *D. polymorpha* ranges between 3–9 years. Maximum growth rates can reach 0.5 mm/day and 1.5–2.0 cm/year. Adults are sexually mature at 8–9 mm in shell length (i.e. within one year in favorable growing conditions).

Zebra mussels are filter feeders having both inhalant and exhalant siphons. They are capable of filtering about one liter of water per day while feeding primarily on algae. Zebra mussels are able to filter particles smaller than 1 μm in diameter, although they preferentially select larger particles (Sprung and Rose 1988). At a 90% efficiency rate, zebra mussels are much more efficient at filtering such small particles than are unionids and Asiatic clams (Noordhuis et al. 1992). Bacteria, which *D. polymorpha* also tends to filter more effectively than native unionids, may represent another important food source (Cotner et al. 1995, Silverman et al. 1996, Silverman et al. 1997). Microzooplankton (e.g., rotifers and veligers) are ingested by zebra mussels, but larger zooplankton are not eaten (MacIsaac et al. 1991, MacIsaac et al. 1995). Veligers also filter material, but their impact is far less than that of sessile adults. Settled mussels exerted 103 times the grazing rate of veligers in western Lake Erie, for example (MacIsaac et al. 1992).

Filtration rate is highly variable, depending on temperature, concentration of suspended matter, phytoplankton abundance, and mussel size (reviewed by Noordhuis et al. 1992). Zebra mussels can adjust their filtration rates (more frequent interruption of filtering or slower pumping rates) and/or produce pseudofeces above an incipient limiting concentration (ILC) of algae to maintain a constant consumption rate (Fanslow et al. 1995, MacMahon 1996, Sprung and Rose 1988). Feeding activity can be described by the clearance rate (percentage of algal biomass removed from the water column over time), biomass of cleared algae (BCA), feces production and pseudofeces production (μg F or P/BCA). For example, Berg et al. (1996) examined the effects of zebra mussel size and algae species and concentration on zebra mussel feeding activity. Clearance rates were constant over varying concentrations of pure cultures of *Chlamydomonas reinhardtii*, a spherical unicellular species 7.42 μm (± 0.13 μm) in diameter. This indicates that the concentrations used in experiments were below the ILC. However, clearance rates decreased, with increasing concentrations of *Pandorina morum*, a species made up of colonies with varying numbers of cells that are individually as large as *C. reinhardtii*. This indicates that the concentrations used in experiments were above the ILC. Large zebra mussels (20-25 mm in length) displayed a higher clearance rate across all concentrations of *C. reinhardtii* than did small mussels (10-15 mm). Incipient limiting concentration differed in this study from previous studies conducted with European populations. Vanderploeg et al. (2017) found that seston quality and availability affects zebra mussel feeding rate and excretion levels. In Saginaw Bay, filtration rate increased with higher seston concentrations and water temperatures, but was not found to be related to seston composition (POC:TSS, chl:TSS) (Fanslow et al. 1995). No diel patterns of filtration rate have been found. During spring, filtration rates rise dramatically as waters warm from 5–10°C, then level off with respect to temperature, and may be inhibited at temperatures over 20°C. Increased suspended matter can reduce filtration activity to a minimum required to maintain oxygen demand. A sigmoidal relationship exists with filtration rate and size, but this may be an effect of aging. Thus, zebra mussel size, phytoplankton species, and regional population differences will affect clearance rates, ILC, and feces/pseudofeces production.

Material filtered by zebra mussels is either ingested or expelled as feces or mucus-covered pseudofeces. True fecal pellets are chemically altered, larger, and denser. Zebra mussels produce pseudofeces to avoid ingesting non-food material. Pseudofeces production may also be a mechanism to deal with overabundance of food (e.g., algal concentrations above the ILC, incipient limiting concentration), and possibly as a way to reject unpalatable algae. Pseudofeces production increases with increasing suspended solid concentration, as well as increasing temperature, albeit to a much lesser extent (MacIsaac and Rocha 1995, Noordhuis et al. 1992).

Means of Introduction: A release of larval mussels during the ballast exchange of a single commercial cargo ship traveling from the north shore of the Black Sea to the Great Lakes has been deduced as the likely vector of introduction to North America (McMahon 1996). Its rapid dispersal throughout the Great Lakes and major river systems was due to the passive drifting of the larval stage (the free-floating or "pelagic" veliger), and its ability to attach to boats navigating these lakes and rivers (see Remarks section below). Its rapid range expansion into connected waterways was probably due to barge traffic where it is theorized that attached mussels were scraped or fell off during routine navigation. Overland dispersal is also a possibility for aiding zebra mussel range expansion. Many small inland lakes near the Great Lakes unconnected by waterways but accessed by individuals trailering their boats from infested waters have populations of zebra mussels living in them. At least nineteen trailered boats crossing into California had zebra mussels attached to their hulls or in motor compartments; all were found during inspections at agricultural inspection stations. Under cool, humid conditions, zebra mussels can stay alive for several days out of water. Migrating blue catfish (*Ictalurus furcatus*) have shown the potential to pass live adults through their gut when the mussel was consumed and digested in cooler water (<21.1° C) (Gatlin et al. 2013). In March 2021, zebra mussels were discovered attached to imported Marimo balls (an alga in the aquarium trade) in some pet stores of 29 states from Alaska to Florida.

Status: Established in all the Great Lakes, all of the large navigable rivers in the eastern United States, and in many small lakes in the Great Lakes region.

Impact of Introduction:

Summary of species impacts derived from literature review. Click on an icon to find out more...

Ecological	Economic	Human Health	Other

Zebra mussels are notorious for their biofouling capabilities by colonizing water supply pipes of hydroelectric and nuclear power plants, public water supply plants, and industrial facilities. They colonize pipes constricting flow, therefore reducing the intake in heat exchangers, condensers, fire fighting equipment, and air conditioning and cooling systems. Zebra mussel densities were as high as 700,000/m^2 at one power plant in Michigan and the diameters of pipes have been reduced by two-thirds at water treatment facilities. Although there is little information on zebra mussels affecting irrigation, farms and golf courses could be likely candidates for infestations. Navigational and recreational boating can be affected by increased drag due to attached mussels. Small mussels can get into engine cooling systems causing overheating and damage. Navigational buoys have been sunk under the weight of attached zebra mussels. Fishing gear can be fouled if left in the water for long periods. Deterioration of dock pilings has increased when they are encrusted with zebra mussels. Continued attachment of zebra mussel can cause corrosion of steel and concrete affecting its structural integrity.

Zebra mussels can have profound effects on the ecosystems they invade. They primarily consume phytoplankton, but other suspended material is filtered from the water column including bacteria, protozoans, zebra mussel veligers, other microzooplankton and silt. Large populations of zebra mussels in the Great Lakes and Hudson River reduced the biomass of phytoplankton significantly following invasion. Diatom abundance declined 82–91% and transparency as measured by Secchi depth increased by 100% during the first years of the invasion in Lake Erie (Holland 1993). As the invasion spread eastward during 1988 to 1990, successive sampling stations recorded declines in total algae abundance from 90% at the most western station to 62% at the most eastern (Nichols and Hopkins 1993). In Saginaw Bay, sampling stations with high zebra mussel populations experienced a 60–70% drop in chlorophyll-a and doubling of Secchi depth (Fahnenstiel et al. 1993). Phytoplankton biomass declined 85% following mussel invasion in the Hudson River (Caraco et al. 1997). The extent of change that zebra mussels can exert on species composition of the phytoplankton community is unresolved. Increased water clarity allows light to penetrate further, potentially promoting macrophyte populations (Skubinna et al. 1995). As macrophytes can be colonized by veligers, the macrophyte community may be altered if such colonization proves detrimental. Increased light penetration may also cause water temperatures to rise and thermoclines to become deeper, but these effects have not yet been documented. As phytoplankton are consumed, the dissolved organic carbon (DOC) concentration may drop. Indeed, inland lakes with zebra mussels have been found to have lower concentrations of DOC (Raikow 2002). Macrophytes could eventually compensate for this since they are also a source of DOC, but there may be a lag period between the time when phytoplankton biomass is down and macrophytes proliferate. This could produce a period of time when UV-B light penetrates deeper into the water column, because DOC absorbs UV-B radiation. Zebra mussels have also recently been shown to be able to directly assimilate DOC (Roditi et al. 2000). Zebra mussels are able to filter particles smaller than 1μm in diameter, although they preferentially select larger particles (Sprung and Rose 1988). Thus bacteria may represent an important food source (Cotner et al. 1995, Silverman et al. 1996). At a 90% efficiency rate, zebra mussels are much more efficient at filtration of such small particles than are unionids and Asiatic clams. Filtering rate is highly variable, depending on temperature, concentration of suspended matter, phytoplankton abundance, and mussel size (reviewed by Noordhuis et al. 1992). Although European zebra mussels are less active in winter, this seasonal pattern is temperature driven. No diel patterns of filtration rate have been found. During spring, filtration rates rise dramatically between 5 and 10oC, then level off with respect to temperature, and may be inhibited at temperatures over 20oC. Increased suspended matter can reduce filtration activity to a minimum required to maintain oxygen demand. A sigmoidal relationship exists with filtration rate and size, but this may be an affect of aging. Material filtered by zebra mussels is either ingested or expelled as feces or mucus covered pseudofeces. True fecal pellets are chemically altered, larger and more dense. Pseudofeces production increases with increasing suspended solid concentration, as well as increasing temperature, albeit to a much lesser extent (Noordhuis et al. 1992, MacIsaac and Rocha 1995). The rate of biosedimentation through pseudofeces production was very high (28mg/cm2 day at a density of 1180 individuals/m2) under turbid conditions in Lake Erie, lending support to the hypothesis that zebra mussels are responsible for increased water clarity observed since mussel introduction (Klerks et al. 1996). Filtration rate was not related to seston composition (POC:TSS, chl:TSS) in Saginaw Bay (Fanslow et al. 1995). Veligers also filter material, but their impact is far less than that of sessile adults. Settled mussels exerted 103 times the grazing rate of veligers in western Lake Erie, for example (MacIsaac et al. 1992). Microzooplankton, e.g. rotifers and veligers, are ingested by zebra mussels, but larger zooplankton are not eaten (MacIsaac et al. 1991, MacIsaac et al. 1995). It has been speculated that benthic deposition of feces and pseudofeces may aid bacterial productivity, thus producing a source culture that zebra mussels can feed upon (Silverman et al. 1996). It has also been speculated that biodeposition of feces and pseudofeces might cause observed increases in benthic macroinvertebrate populations (Stewart and Haynes 1994).

Biomagnification of Polychlorinated Biphenyls (PCBs) was observed in Gammarus associated with zebra mussels, indicating concentration of pollutants in zebra mussel feces or pseudofeces can transfer to other trophic levels (Bruner et al. 1994). In an experimental study, however, Botts et al. (1996) found greater abundances of macroinvertebrates associated with both living and non-living (i.e. empty shell) zebra mussel druses compared with their no-druse treatment. Thus the increased physical habitat complexity of a mussel colony may benefit macroinvertebrates rather than deposition of feces and pseudofeces. Zebra mussels can reduce filtration rates (more frequent interruption of filtering or slower pumping rates) and/or produce pseudofeces above an incipient limiting concentration (ILC) of algae to maintain a constant consumption rate (Sprung and Rose 1988, Fanslow et al. 1995, MacMahon 1996). Feeding activity can be described by the clearance rate (percentage of algal biomass removed from the water column over time), biomass of cleared algae (BCA), feces production and pseudofeces production (μg F or P/BCA). For example, Berg et al. (1996) examined the effects of zebra mussel size and algae species and concentration on zebra mussel feeding activity. Clearance rates were constant over varying concentrations of pure cultures of *Chlamydomonas reinhardtii*, a spherical unicellular species of 7.42 μm (± 0.13μm) in diameter. This indicates that the concentrations used in experiments were below the ILC. However, clearance rates decreased, with increasing concentrations of Pandorina morum, a species made up of colonies with varying numbers of cells that are individually as large as *C. reinhardtii*. This indicates that the concentrations used in experiments were above the ILC. Large zebra mussels (20-25 mm in length) displayed a higher clearance rate across all concentrations of *C. reinhardtii* than did small mussels (10-15 mm). Incipient limiting concentration differed in this study from previous studies done with European populations. Thus zebra mussel size, phytoplankton species, and regional population differences affect clearance rates, ILC and feces/pseudofeces production. Zebra mussels produce pseudofeces to avoid ingesting non-food material (e.g. clay), as a mechanism to deal with overabundance of food (e.g. algal concentrations above the ILC), and possibly as a way to reject unpalatable algae. Zebra mussels readily reject blue-green algae, such as Microcystis, as pseudofeces (Vanderploeg et al. 2001). The presence of this cyanobacterium does not inhibit filtering, except in mass abundances such as a bloom (Noordhuis et al. 1992, Lavrentyev et al. 1995). Zebra mussels can select material for rejection through pseudofeces production internally, perhaps identifying cyanobacteria by chemical cues (ten Winkel and Davids 1982). Inland lakes with lower nutrient levels have been observed to be more frequently dominated by Microcystis when invaded by zebra mussels (Raikow et al. 2004). Understanding of the fate of pseudofeces once it expelled is poor. Zebra mussels removed metals from the water column of Lake Erie and deposited it to the bottom at high rates (Klerks et al. 1996). Roditi et al. (1997) found that the biodeposits of zebra mussel were organically enriched, including 3.9% live algae by weight. Resuspension of this material occurred in their system, a tidal estuary, reducing the potential impact of biodeposition to the benthos. Less well known is the fate of live algae bound into pseudofeces. Bastviken et al. (1998) speculate that phytoplankton which survives the pseudofeces process must be resuspended in order for long term survival, a process less likely to occur in inland lakes than in tidal estuaries. If survivorship following filtration is equal between phytoplankton species, then community species composition can remain unchanged. Other factors may affect the phytoplankton community, however, including increased light.

The zooplankton community has also been affected by the invasion of zebra mussels. Zooplankton abundance dropped 55-71% following mussel invasion in Lake Erie, with microzooplankton more heavily impacted (MacIsaac et al. 1995). Mean summer biomass of zooplankton decreased from 130 to 78 mg dry wt. m-3 between 1991 and 1992 in the inner portion of Saginaw Bay. The total biomass of zooplankton in the Hudson River declined 70% following mussel invasion, due both to a reduction in large zooplankton body size and reduction in microzooplankton abundance. These effects can be attributed to reduction of available food (phytoplankton) and direct predation on microzooplankton. Increased competition in the zooplankton community for newly limited food should result from zebra mussel infestation. The size of individual zooplankters might decrease. Hypotheses can be formulated specifying which species will prevail based on knowledge of competitive ability.

Effects may continue through the food web to fish. Reductions in zooplankton biomass may cause increased competition, decreased survival and decreased biomass of planktivorous fish. Alternatively, because microzooplankton are more heavily impacted by zebra mussels the larval fish population may be more greatly affected than later life stages. This may be especially important to inland lakes with populations of pelagic larval fish such as bluegills. Benthic feeding fish may benefit as opposed to planktivorous fish, or behavioral shifts from pelagic to benthic-feeding may occur. In addition, proliferation of macrophytes may alter fish habitat. Experimental evidence exists that zebra mussels can reduce the growth rate of larval fish through food web interactions (Raikow 2004). Conclusive negative impacts on natural populations of fish, however, have yet to observed (see Raikow 2004). Other effects include the extirpation of native unionid clams through epizootic colonization (Schloesser et al. 1996, Baker and Hornbach 1997). Zebra mussels restrict valve operation, cause shell deformity, smother siphons, compete for food, impair movement and deposit metabolic waste onto unionid clams. Survival rates of native unionid mussels in the Mississippi River, Minnesota have been shown to decline significantly with the increase in zebra mussel colonization (Hart et al. 2001).To date, unionids have been extirpated from Lake St. Clair and nearly so in western Lake Erie. Many species of birds known to be predators of zebra mussels occur in the Great Lakes region. While a new food source may benefit such predators, biomagnification of toxins into both fish and birds is possible. Some effects have been hypothesized as worst-case scenarios. For example, zebra mussels may cause a shift from pelagically to benthically-based food webs in inland lakes. Zebra mussels may also shift lakes from a turbid and phytoplankton-dominated state to clear and macrophyte-dominated state, i.e. between alternative stable equilibria (Scheffer et al. 1993).

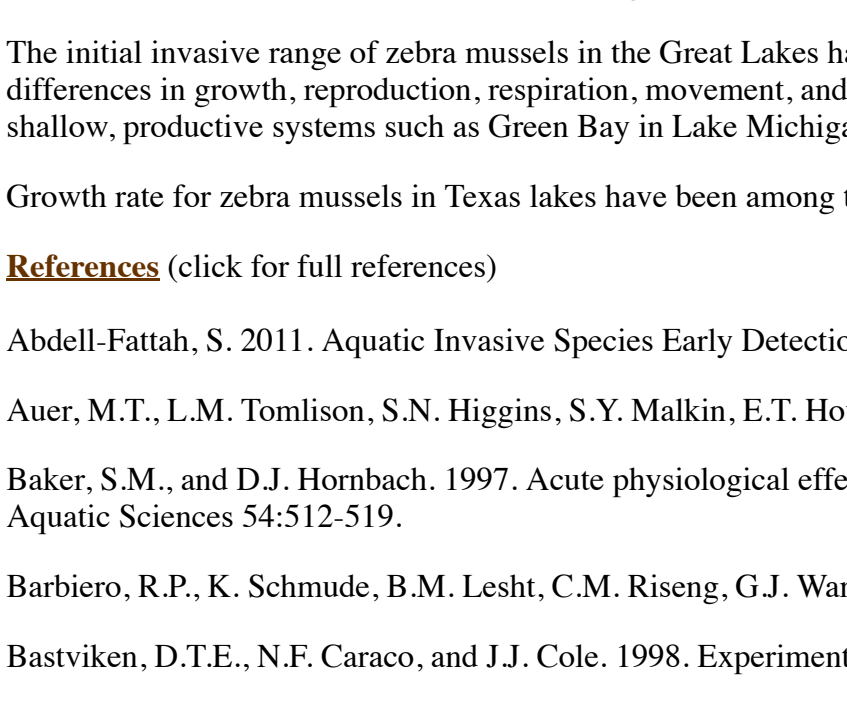


Image by K. Holcomb

Remarks: Zebra mussels represent one of the most important biological invasions into North America, having profoundly affected the science of invasion biology along with public perception and policy. In the 1980s, invasion biology began to emerge as a true sub-discipline of ecology as evidenced by an exponential increase in scientific output on the subject (Raikow, unpubl. data). Most work on the subject was terrestrial. Invasions were not a large component of the popular environmental movement, and no serious legislation existed concerning invasions beyond agricultural pests. After the discovery of zebra mussels in 1988, the exponential rate of scientific output on invasions itself increased (Raikow, unpubl. data), the Nonindigenous Aquatic Nuisance Prevention and Control Act was written and passed, and invasions became a topic discussed in the media. Today, biological invasions are described as the second leading cause of extinction behind habitat destruction. Aquatic invasions are a topic of much research. For these reasons the zebra mussel is often described as the "poster child" of biological invasions.

A long tradition of zebra mussel study exists in Europe and the former Soviet Union, where the zebra mussel has been present for 150 years (see Mackie et al. 1989 for an annotated bibliography of European references). Work includes spatial distribution patterns, demography, tolerance limits for physical and chemical parameters, and physiology. Extensive ecological work in the United States began soon as the zebra mussel was discovered and peaked in the early 1990s. The literature on ecosystem and community-level effects of zebra mussels has been dominated by work investigating Lake Erie, Saginaw Bay, the Hudson River, and Oneida Lake (e.g., Fahnenstiel et al.1993, Holland 1993, Pace et al. 1998, Idrisi et al. 2001.

The rapid invasion of North American waterways has been facilitated by the zebra mussel's ability to disperse during all life stages. Passive drift of large numbers of pelagic larval veligers allows invasion downstream. Yearlings are able to detach and drift for short distances. Adults routinely attach to boat hulls and floating objects and are thus anthropogenically transported to new locations. Transporting recreational boats disperses zebra mussels between inland lakes. In addition, speculation exists that waterfowl can disperse zebra mussels, but this has yet to be conclusively demonstrated. While byssal threads develop in the larvae of some non-dreissenid endemic bivalves and are used to attach to fish gills, there are no endemic freshwater bivalves with byssal adult stages. This adaptation has been important to the zebra mussel's success in invading North America.

The initial invasive range of zebra mussels in the Great Lakes has decreased due to displacement by the congeneric quagga mussel. There are multiple mechanisms by which quagga mussels displace zebra mussels, including differences in growth, reproduction, respiration, movement, and development (Ram et al. 2012; Karateyev et al. 2015; D'Hont et al. 2021). Zebra mussels still dominate in inland lakes and rivers and the two species coexist in shallow, productive systems such as Green Bay in Lake Michigan, Saginaw Bay in Lake Huron, and Western Lake Erie.

Growth rate for zebra mussels in Texas lakes have been among the fastest recorded in North American and Europe, at 127.9 µm day^{-1} (Locklin et al. 2020).

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Other Resources:
Current Printable Zebra and Quagga Mussel Sightings Distribution Map (PNG) (PDF)

Zebra Mussel Images (Public Domain)

USGS Quagga Mussel Fact Sheet

USGS *Dreissena* species Frequently Asked Questions

USGS Zebra Mussel Frequently Asked Questions

Invasive Mussel Collaborative

- Dreissena Project Coordination Mapper

Student Questions Concerning Zebra and Quagga Mussels

Dreissena species (New York Invasive Species Information)

Dreissena polymorpha (Global Invasive Species Database)

Great Lakes Waterlife

US Fish and Wildlife Service Ecological Risk Screening Summary for *Dreissena polymorpha*

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